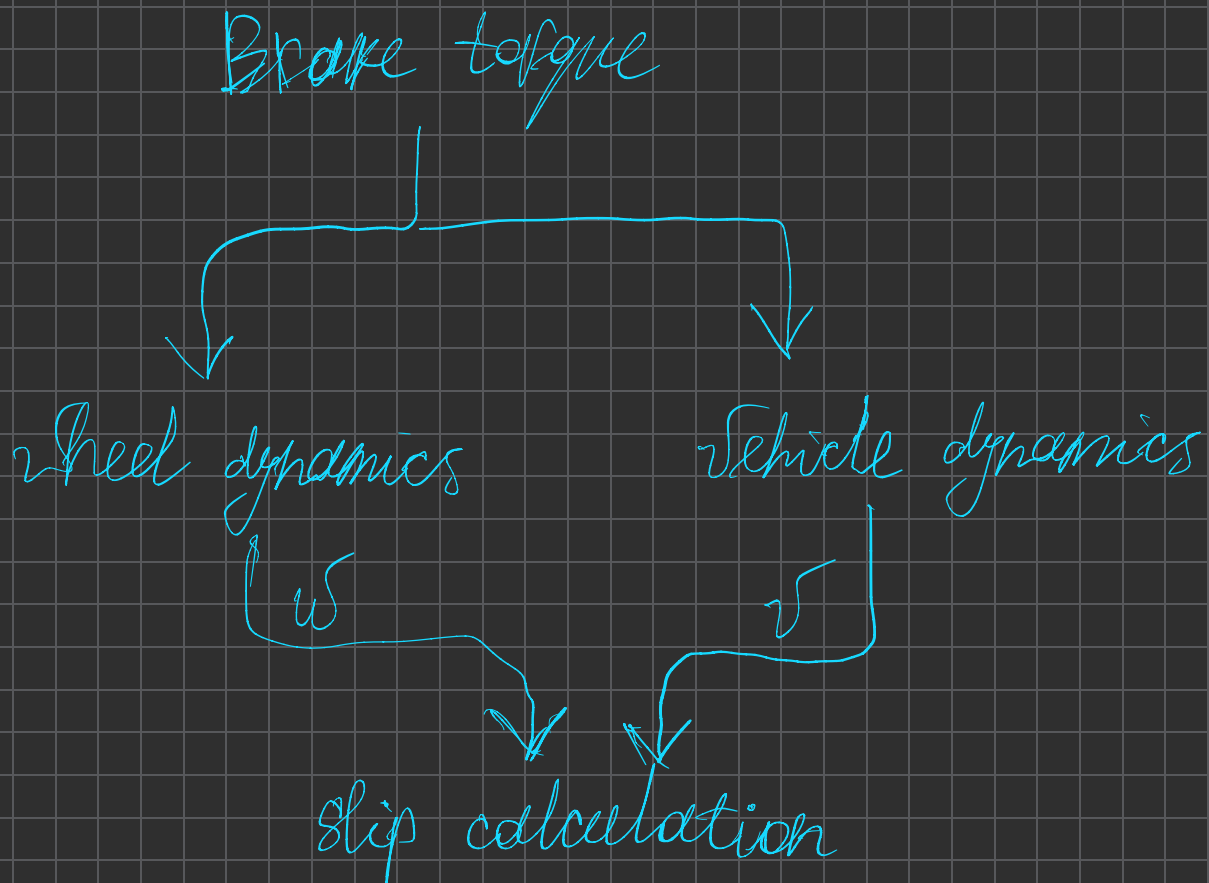
$$s = \frac{v - \omega R}{v}$$

$$F_x = \mu(s) F_N \quad \text{one wheel}$$

$$m \frac{dv}{dt} = -F_x$$

$$J \frac{d\omega}{dt} = F_x R - T_b$$



wheel might start down faster than vehicle

$v(t)$

$$\Sigma F = ma$$

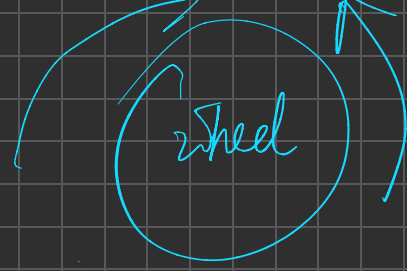
$$a = \frac{-\Sigma F}{m} \rightarrow \text{sum of braking forces}$$

$$\frac{dv}{dt} = \frac{-\Sigma F}{m} \qquad \Sigma F = -F_x = -\mu N$$

$$v = \int \frac{\Sigma F}{m}\, dt$$

$\omega(t)$

Brakes



friction

$$\tau = F_B R, \quad F_B = \mu N$$

$$v(t) = v(0) - \int_0^t \frac{\Sigma F}{m}\, dt$$

$25\ m\,s^{-1}$

$$J\frac{d\omega}{dt} = F_x R - T_B$$

$$\frac{d\omega}{dt} = \frac{1}{J}\left(F_x R - T_B\right)$$

$$\omega(t) = \omega_0 + \frac{1}{J}\int \left(F_x R - T_B\right) dt$$

$$v_0 = R\omega_0 \Rightarrow \omega_0 = \frac{v_0}{R}$$

5

$$s(t) = \frac{v(t) - w(t)R}{v(t)}$$

$$\frac{s - s_1}{s_2 - s_1} = \frac{\mu - \mu_1}{\mu_2 - \mu_1}$$

↳ proceeding to Simulink

assume $s = 0 \Rightarrow \mu = 0 \Rightarrow F_\mu = 0$

$$w(t) = w(0) - \frac{1}{J_0}\int_0^t T_B \, dt$$

T_B - constant

J - constant

$$w(t) = w(0) - \frac{T_B}{J} * t$$